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United States Patent Application Publication

20250271813

Kind Code

A1

Publication Date

August 28, 2025

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INTERNAL CHIME MECHANISM FOR AN ALARM CLOCK

Abstract

An electromechanical chime mechanism includes a chime bar having a plurality of rectangular undercuts and a solenoid striker having a mallet head positioned proximate to and substantially perpendicular to the chime bar. The solenoid striker is (i) positioned within an acoustic resonator and (ii) configured to propel the mallet head against the chime bar when electrically activated. The chime mechanism includes a plurality of shock-absorbing elastic posts with crescent tops holding the chime bar in place. The acoustic resonator is a cavity positioned behind the chime bar with an opening towards the chime bar.

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Family ID: 1000008361648

Appl. No.: 18/989574

Filed: December 20, 2024

Related U.S. Application Data

us-provisional-application US 63557464 20240223

Publication Classification

Int. Cl.: G04C21/18 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/557,464, filed Feb. 23, 2024, titled “Internal Chime Mechanism for an Alarm Clock,” which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This application relates generally to chime mechanisms, and more specifically to chime mechanisms within limited enclosures.

BACKGROUND

[0003] Many alarm clocks use sounds to wake people from sleep. Typically, the sounds are electronically generated, making loud beeps or buzzing sounds.

SUMMARY

[0004] Disclosed is an alarm clock design featuring a novel compact electromechanical chime mechanism. The chime mechanism includes a rounded rectangular chime bar with a unique cross-sectional design, a solenoid striker with a mallet head, shock-absorbing rubber mounts, and an acoustic resonator configured in a novel arrangement to enhance sound resonance and clarity in a very compact size.

[0005] In accordance with some embodiments, an electromechanical chime mechanism includes a chime bar having a plurality of rectangular undercuts (e.g., two or more undercuts) and a solenoid striker having a mallet head positioned proximate to and substantially perpendicular to the chime bar. The solenoid striker is (i) positioned within an acoustic resonator and (ii) configured to propel the mallet head against the chime bar when electrically activated. The acoustic resonator is a cavity positioned behind the chime bar with an opening towards the chime bar.

[0006] In accordance with some embodiments, a rectangular chime bar includes a plurality of rectangular undercuts. The chime bar is configured to output sound at a predetermined frequency in response to being struck. A fundamental vibrational mode of the chime bar corresponds to the predetermined frequency. A first position along the longer dimension of the chime bar corresponds to a first node of a fundamental frequency and a second position along the longer dimension of the chime bar corresponds to a second node of the fundamental frequency. The plurality of rectangular undercuts includes a first undercut located at the first position and a second undercut located at the second position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a better understanding of the various embodiments, review the Description of Embodiments below together with the following drawings. When a reference numeral is used in multiple drawings, it represents the same element in each place it is used.

[0008] FIG. 1A is a perspective exterior view of an alarm clock in accordance with some embodiments.

[0009] FIG. 1B is the same perspective view of the alarm clock as shown in FIG. 1A, but with exterior portions of the alarm clock shown as transparent, in accordance with some embodiments.

[0010] FIG. 1C is a close-up perspective view of the alarm clock shown in FIG. 1A, with the front face removed and other exterior portions shown as transparent, in accordance with some embodiments.

[0011] FIG. 1D is a top view of the alarm clock shown in FIG. 1A, with the exterior portion shown as transparent, in accordance with some embodiments.

[0012] FIG. 1E is a side view of the alarm clock shown in FIG. 1A, with exterior portions shown as transparent, in accordance with some embodiments.

[0013] FIG. 2A is the same perspective view of the alarm clock as shown in FIG. 1B, with some components removed to visualize the chime bar and shock absorbing mounts inside the alarm clock, in accordance with some embodiments.

[0014] FIG. 2B is the same top view of the alarm clock as shown in as FIG. 1D, with some components removed to visualize the chime bar, solenoid striker, and shock absorbing mounts inside the alarm clock, in accordance with some embodiments.

[0015] FIG. 2C is the same side view of the alarm clock as shown in as FIG. 1E, with some components removed to visualize the chime bar, solenoid striker, and shock absorbing mounts inside the alarm clock, in accordance with some embodiments.

[0016] FIG. 3A is a perspective view of the chime bar, in accordance with some embodiments.

[0017] FIG. 3B is a side view of the chime bar shown in FIG. 3A, in accordance with some embodiments.

[0018] FIG. 3C is a front view of the chime bar shown in FIG. 3A, in accordance with some embodiments.

[0019] FIG. 3D illustrates a fundamental mode of the chime bar, in accordance with some embodiments.

[0020] FIG. 4A is a perspective view of a mounting post and shock absorbing mounts, in accordance with some embodiments.

[0021] FIG. 4B is a front view of the mounting post and shock absorbing mounts shown in FIG. 4A, in accordance with some embodiments.

[0022] FIG. 4C is a back view of the mounting post and shock absorbing mounts shown in FIG. 4A, in accordance with some embodiments.

[0023] FIG. 4D is a side view of the mounting post and shock absorbing mounts shown in FIG. 4A, in accordance with some embodiments.

[0024] FIG. 5 illustrates a position at which the chime bar shown in FIG. 3A is struck, in accordance with some embodiments.

DESCRIPTION OF EMBODIMENTS

[0025] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, the various described embodiments may be practiced without these specific details. In some instances, well-known methods, procedures, and/or components have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0026] The terms “first” and “second” may be used to distinguish one element from another, without implying order. For example, a first contact could be termed a second contact or a second contact could be termed a first contact. The first contact and the second contact are both contacts, but they are not the same contact, unless the context clearly indicates otherwise.

[0027] The terminology used here describes particular embodiments and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. The terms “includes,” “including,” “comprises,” and/or “comprising,” specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0028] While certain materials are sometimes specified, the disclosed devices are not limited to these example materials. Plastics, rubbers, metals, woods, and other materials may comprise some

or all of the elements of the various embodiments.

[0029] FIG. 1A is a perspective exterior view of an alarm clock, in accordance with some embodiments. FIGS. 1B-1E show the alarm clock **100**, but with exterior portions of the alarm clock shown as transparent. FIGS. 1B and 1C are perspective views of the alarm clock **100**, FIG. 1D is a top view of the alarm clock **100**, and FIG. 1E is a side view of the alarm clock **100**.

[0030] As shown in FIGS. 1B-1E, the alarm clock **100** includes a chime bar **300** configured to emit (e.g., output or produce) a sound (e.g., ring or chime) at a predetermined frequency when struck. In some embodiments, the chime bar **300** (illustrated in FIGS. 1D and 1E with medium shading for ease of reference) is struck with a mallet head **206** (illustrated in FIGS. 1D and 1E with dark shading for ease of reference) that is propelled by a solenoid striker **204**. The chime bar **300** is configured to emit sound in response to being struck by the mallet head **206**, providing an alarm sound for the alarm clock **100**. Details regarding the components of the alarm clock **100** and their functionality are described below with respect to FIGS. 2A-2C.

[0031] FIGS. 2A-2C illustrate the alarm clock **100**, but with exterior portions of the alarm clock shown as transparent and with some components removed so that the chime bar **300**, the solenoid striker **204**, and the mounting apparatus **400** (as shown in FIG. 4A, including the mounting post **402** and the shock absorbing mounts **404**) can be clearly visualized. FIG. 2A shows the same perspective view as FIG. 1B, FIG. 2B shows the same top view as FIG. 1D, and FIG. 2C shows the same side view as FIG. 1E.

[0032] The alarm clock **100** includes an acoustic resonator **202** (also referred to herein as a resonator **202**), a solenoid striker **204**, a chime bar **300**, and a mounting apparatus **400** (see, e.g., FIG. 4A) for the chime bar **300**. The chime bar **300** (illustrated in FIGS. 2B and 2C with medium shading for ease of reference) is configured to emit a sound (e.g., ring or chime) at a predetermined frequency when struck. Details regarding the chime bar **300** are provided below with respect to FIGS. 3A-3D. The mounting apparatus **400** is configured to maintain the chime bar's position while ensuring that its natural vibration is not hindered. A mounting post **402** of the mounting apparatus **400** is illustrated in FIGS. 2B and 2C with light shading for ease of reference. Details regarding the mounting apparatus **400** are provided below with respect to FIGS. 4A-4D. The solenoid striker **204** is configured to propel the mallet head **206** (illustrated in FIGS. 2B and 2C with dark shading for ease of reference) to strike the chime bar **300**, and the resonator **202** is configured to amplify sounds emitted from the chime bar **300**.

[0033] In some embodiments, as shown in FIGS. 1B-1E and 2A-2C, the resonator **202** is an acoustic cavity positioned behind the chime bar **300** with an opening towards the chime bar **300**. This design allows sound waves generated by the chime bar **300** to enter and resonate within the cavity, amplifying the sound in a manner that highlights the desired musical notes while dampening unwanted vibrations.

[0034] In some embodiments, as shown in FIGS. 1B-1E and 2A-2C, the solenoid striker **204** (also referred to as a solenoid mechanism or solenoid) is positioned within the acoustic resonator **202**. The solenoid striker **204** utilizes an electrically activated solenoid **204** to propel a mallet head **206** against the chime bar **300**. In some embodiments, the mallet head **206** is positioned proximate to and substantially perpendicular (e.g., ± 3 degrees, ± 4 degrees, ± 5 degrees, ± 10 degrees, or ± 15 degrees) to the chime bar **300**.

[0035] in

[0036] This unique configuration contrasts with conventional mechanisms where the striker typically impacts the chime bar **300** from an external position relative to the resonator **202**. This internal placement is partially motivated by the compact design requirements of the alarm clock **100**. Housing the solenoid striker **204** within the resonator **202** introduces unique acoustic challenges. Among the challenges are the amplification of mechanical noise generated by the solenoid striker **204** itself, which muddles the clarity of the chime sound output from the chime bar **300**. To mitigate the unintended amplification of mechanical noise, the solenoid striker **204** is

mounted on specialized stops **208**, which are made of an elastic material, such as rubber or similar material. These stops **208** act as vibration dampeners, absorbing and minimizing the transmission of vibrational energy from the solenoid striker **204** to the resonator **202**. Additionally, viscous grease is applied to the actuator of the solenoid striker **204**. This grease serves a dual purpose: it lubricates moving parts to reduce friction and wear, and it dampens sounds (e.g., mechanical sounds) created by the actuator's movement.

[0037] FIGS. **3A-3D** show the chime bar **300**, in accordance with some embodiments. FIG. **3A** is a perspective view of the chime bar **300**, FIG. **3B** is a side view of the chime bar **300**, and FIG. **3C** is a front view of the chime bar **300**. FIG. **3D** illustrates a fundamental mode of the chime bar **300**.

[0038] The chime bar **300** is configured to emit sound at a predetermined frequency that is lower than the frequency output from a musical bar composed of the same material as the chime bar **300** and having the same outer dimensions as the chime bar **300**. This is accomplished by the use of one or more undercuts **308**. The chime bar **300** may include any number of undercuts **308**. The undercuts **308** are most easily seen in FIGS. **3A** and **3B**.

[0039] In some embodiments, such as shown in FIGS. **3A-3D**, the chime bar **300** possesses a unique cross section incorporating three undercuts **308-1**, **308-2**, and **308-3** (also referred to collectively as undercuts **308**). This design allows the chime bar **300** to vibrate at a desired musical frequency while having a significantly smaller size than conventional musical bars of the same frequency. In some embodiments, there are fewer or more undercuts. In some embodiments, the undercuts **308** are square. In some embodiments, the undercuts **308** are rectangular without being square. In some embodiments, the undercuts have edges that are nonlinear (e.g., circular, elliptical, or parabolic) or having portions that are nonlinear.

[0040] FIG. **3D** illustrates a position **300-A** (shown as a thick dashed line) of the chime bar **300** when it is oscillating at the fundamental mode (e.g., the fundamental vibrational mode, sometimes annotated as $n=1$ or $f_{\text{sub.1}}$) of the chime bar **300**. The fundamental mode determines the frequency of sound produced by the chime bar **300**, and the fundamental mode of the chime bar **300** depends on: (i) the density and (ii) the elasticity of the material from which the chime bar **300** is made, as well as (iii) the length (e.g., long dimension) of the chime bar **300**. As shown in FIG. **3D**, the fundamental mode has two nodes **310-1** and **310-2**, which correspond to positions along the length of the chime bar **300** that do not move (e.g., have an amplitude of zero) during oscillation of the chime bar **300**. In some embodiments, the nodes are located at substantially the center (in the short direction, H) of the chime bar **300**.

[0041] In some embodiments, the chime bar **300** is made of aluminum (e.g., Aluminum 6063). In some embodiments, the chime bar **300** is made of a material having a density that is substantially 2.7 grams/centimeter.sup.3 (g/cm.sup.3) (e.g., ± 0.3 g/cm.sup.3) and having an elasticity modulus that is substantially 68.9 GigaPascals (GPa) (e.g., ± 1 GPa). In some embodiments, the chime bar **300** has a length L (e.g., the long dimension) that is no greater than 147 mm, a height H (e.g., the short dimension) that is no greater than 38 mm, and a width W (e.g., the thickness) that is no greater than 3.1 mm. An equivalent musical bar (e.g., constructed out of the same material and having the same outer dimensions) that does not include undercuts (e.g., a solid bar with no undercuts) would produce a sound that has a higher frequency than the predetermined frequency of the chime bar **300**. In some embodiments, the chime bar **300** is configured to produce a B4 note (e.g., the middle B note) having a frequency of 493.883 Hz ± 10 cents ("cents" is the well-known logarithmic unit of measurement to describe musical intervals). In some embodiments, the chime bar **300** is configured to produce sound having a frequency within the range of 491 Hz-495 Hz.

[0042] For example, an equivalent musical bar (e.g., constructed out of the same material and having the same outer dimensions) that does not include undercuts (e.g., a solid bar with no undercuts) would produce sound around 790 Hz ± 10 cents.

[0043] In another example, a musical bar composed of the same material as the chime bar **300** and having no undercuts has outer dimensions of approximately 45 cm $\times$ 3.5 cm $\times$ 3 mm ($L\times H\times W$) to

produce the B4 note (e.g., to produce sound around 493.883 Hz or to produce sound within the range of 491 Hz-495 Hz). The chime bar **300** has a significantly reduced length in comparison to the length of this example musical bar ($L < 15$ cm for the chime bar **300** compared to $L \sim 45$ cm for the example musical bar with no undercuts).

[0044] In some embodiments, the undercuts **308** are located at predetermined positions. In some embodiments, as shown in FIG. 3D, a first undercut **308-1** of the chime bar **300** is located at the first node **310-1** of the fundamental mode of the chime bar **300** (e.g., at least a portion of the first undercut **308-1** overlaps with the position of the first node **310-1**), and a second undercut **308-2** of the chime bar **300** is located at a second node **310-2** of the fundamental mode of the chime bar **300** (e.g., at least a portion of the second undercut **308-2** overlaps with the position of the second node **310-2**). In some embodiments, the chime bar **300** includes a third undercut **308-3** that is located at a node of the second harmonic of the chime bar **300**. The second harmonic of the chime bar **300** contributes to the sound (e.g., frequencies) produced by the chime bar **300**. The second harmonic of the chime bar **300** depends on the fundamental mode of the chime bar **300**. Thus, it also depends on the density and elasticity of the material from which the chime bar **300** is made, as well as the length L (e.g., the long dimension) of the chime bar **300**.

[0045] In some embodiments, the chime bar **300** includes two mounting holes configured to allow the chime bar **300** to be positioned by the mounting apparatus **400**. In some embodiments, the first mounting hole **312-1** is located at the first undercut **308-1** and the second mounting hole **312-2** is located at the second undercut **308-2**. The mounting holes can be seen in FIGS. 3A and 3C. In some embodiments, the first mounting hole **312-1** is located at the first node **310-1** of the fundamental mode of the chime bar **300** (e.g., at least a portion of the first mounting hole **312-1** overlaps with the position of the first node **310-1**), and the second mounting hole **312-2** is located at the second node **310-2** of the fundamental mode of the chime bar **300** (e.g., at least a portion of the second mounting hole **312-2** overlaps with the position of the second node **310-2**).

[0046] FIGS. 4A-4D show a mounting apparatus **400**, which includes a mounting post **402** and shock absorbing mounts **404**, in accordance with some embodiments. FIG. 4A is a perspective view of the mounting post **402** and shock absorbing mounts **404**, FIG. 4B is a front view of the mounting post and shock absorbing mounts, and FIG. 4C is a back view of the mounting post and shock absorbing mounts. FIG. 4D is a side view of the mounting post **402** and shock absorbing mounts **404**, and illustrates the position of the mounting post **402** and shock absorbing mounts **404** relative to the chime bar **300** (shown in FIG. 4D as a dashed rectangle).

[0047] The mounting apparatus **400** includes a plurality of shock absorbing mounts **404** that are fixed to the mounting post **402** (shown in FIGS. 4A-4C with shading). The shock absorbing mounts **404** are constructed of (e.g., made of, composed of) an elastic material such as rubber or a similar material. To position the chime bar **300**, a first shock absorbing mount of the mounting apparatus **400** is seated in the first mounting hole **312-1** (shown in FIG. 3A) and a second shock absorbing mount is seated in the second mounting hole **312-2** (shown in FIG. 3A). Thus, the chime bar **300** is mounted via the shock-absorbing mounts **404**. These shock-absorbing mounts **404** are specifically engineered to tightly control the chime bar's position while ensuring that its natural vibration is not hindered. In some embodiments, the shock-absorbing mounts **404** are precisely positioned at the chime bar's nodes **310-1** and **310-2** (shown in FIG. 3D), ensuring that they do not interfere with the chime bar's natural vibrations. Upon the chime bar **300** being struck, the crescent moon geometry of the upper shock-absorbing mounts **404** allows the shock-absorbing mounts **404** to compress slightly, letting the chime bar **300** move in response to the impact. This movement creates a clearer sound due to the more elastic nature of the collision compared to having a chime bar **300** mounted using a more rigid material. After the impact at the chime bar **300**, the elastic properties of the material of the shock-absorbing mounts **404** (e.g., rubber or a similar material) ensure that the chime bar **300** returns to its original position, ready for the next strike. This recovery is important: if the chime bar **300** were to remain displaced, subsequent strikes by the mallet head **206** (of the

solenoid striker **204**) would fail to connect with the chime bar **300**, compromising the alarm's effectiveness. In some embodiments, the mounting post **402** is constructed from a material different from the shock absorbing mounts **404** (e.g., the mounting post is constructed from a rigid material in some embodiments).

[0048] FIG. 5 illustrates the position **502** at which the chime bar **300** shown in FIG. 3A is struck by the mallet head **206**, in accordance with some embodiments. The solenoid striker **204** and mallet head **206** are precisely positioned so that the mallet head **206** contacts (e.g., strikes, hits) the chime bar **300** at substantially the center **502** of the chime bar **300** (e.g., ± 2 mm along the short direction of the chime bar **300** and ± 2 mm along the long direction of the chime bar **300**). The center **502** (e.g., geometric center) of the chime bar **300** is shown in FIG. 5 as a block dot. Striking the chime bar **300** at an off centered position may excite torsional vibration modes that result in reduced quality in the sound produced by the chime bar **300**. In some embodiments, the mallet head **206** strikes the chime bar **300** on the side of the chime bar **300** that does not include the undercuts **308**. Thus, FIG. 5 shows the position of the undercuts **308** as dotted lines since the undercuts are located on the opposite side of the chime bar **300** as shown in FIG. 5.

[0049] Turning now to some embodiments.

[0050] (A1) An electromechanical chime mechanism includes a chime bar having a plurality of rectangular undercuts and a solenoid striker. The solenoid striker has a mallet head that is positioned proximate to and substantially perpendicular to the chime bar. The solenoid striker is (i) positioned within an acoustic resonator and (ii) configured to propel the mallet head against the chime bar when electrically activated. The acoustic resonator is a cavity positioned behind the chime bar with an opening towards the chime bar.

[0051] (A2) The electromechanical chime mechanism of A1, where the chime bar has a first node and a second node that is distinct from the first node. The first node and the second node correspond to a fundamental vibrational mode of the chime bar. The plurality of rectangular undercuts includes a first rectangular undercut and a second rectangular undercut that is distinct from the first rectangular undercut. The first rectangular undercut is located at a first position that overlaps with the first node and the second rectangular undercut is located at a second position that overlaps with the second node.

[0052] (A3) The electromechanical chime mechanism of A2, where a plurality of shock-absorbing elastic posts have crescent tops holding the chime bar in place. The plurality of shock-absorbing elastic posts includes a first shock absorbing post and a second shock absorbing post that is distinct from the first shock absorbing post. The first shock absorbing post is positioned at the first position and the second shock absorbing post is positioned at the second position.

[0053] (A4) The electromechanical chime mechanism of any of A1-A3, where the mallet head is positioned relative to the chime bar such that the mallet head contacts the chime bar substantially at a center along a short dimension of the chime bar.

[0054] (A5) The electromechanical chime mechanism of any of A1-A4, where the mallet head is positioned relative to the chime bar such that the mallet head contacts the chime bar substantially at a center along a long dimension of the chime bar.

[0055] (A6) The electromechanical chime mechanism of any of A1-A5, where the chime bar is configured to, when struck by the mallet head, output sound at a predetermined frequency that is lower than a natural frequency of a bar that (i) is composed of a material having a substantially similar density and a substantially similar elasticity as a material of the chime bar and (ii) has the same outer dimensions as the chime bar.

[0056] (A7) The electromechanical chime mechanism of any of A1-A6, where the chime bar has outer dimensions that are no greater than 3.1 mm in thickness, 147 mm in length, and 38 mm in width.

[0057] (A8) The electromechanical chime mechanism of any of A1-A7, where the chime bar is composed of a material having a density no greater than 2.8 g/cm³.

[0058] (A9) The electromechanical chime mechanism of any of A6-A8, where the predetermined frequency substantially corresponds to a B4 note.

[0059] (A10) The electromechanical chime mechanism of any of A1-A9, further including a plurality of shock-absorbing elastic posts with crescent tops holding the chime bar in place.

[0060] (A11) The electromechanical chime mechanism of A10, where the shock-absorbing elastic posts are configured to hold the chime bar in place while allowing the chime bar to vibrate freely.

[0061] (A12) The electromechanical chime mechanism of any of A1-A11, where the chime bar has a rounded rectangular shape.

[0062] (B1) A rectangular chime bar includes a plurality of rectangular undercuts. The chime bar is configured to output sound at a predetermined frequency in response to being struck. A fundamental vibrational mode of the chime bar corresponds to the predetermined frequency. A first position along a long dimension of the chime bar corresponds to a first node of a fundamental frequency and a second position along the long dimension of the chime bar corresponds to a second node of the fundamental frequency and the plurality of rectangular undercuts includes a first undercut located at the first position and a second undercut located at the second position.

[0063] (B2) The rectangular chime bar of B1, where the predetermined frequency is lower than a natural frequency of a bar composed of a same material and having same outer dimensions as the chime bar.

[0064] The illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best utilize the invention and various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. An electromechanical chime mechanism comprising: a chime bar having a plurality of rectangular undercuts; and a solenoid striker having a mallet head positioned proximate to and substantially perpendicular to the chime bar, wherein the solenoid striker is (i) positioned within an acoustic resonator and (ii) configured to propel the mallet head against the chime bar when electrically activated, and wherein the acoustic resonator is a cavity positioned behind the chime bar with an opening towards the chime bar.
2. The electromechanical chime mechanism of claim 1, wherein: the chime bar has a first node and a second node that is distinct from the first node; the first node and the second node correspond to a fundamental vibrational mode of the chime bar; the plurality of rectangular undercuts includes a first rectangular undercut and a second rectangular undercut that is distinct from the first rectangular undercut; the first rectangular undercut is located at a first position that overlaps with the first node; and the second rectangular undercut is located at a second position that overlaps with the second node.
3. The electromechanical chime mechanism of claim 2, further comprising: a plurality of shock-absorbing elastic posts with crescent tops holding the chime bar in place, wherein: the plurality of shock-absorbing elastic posts includes a first shock absorbing post and a second shock absorbing post that is distinct from the first shock absorbing post; the first shock absorbing post is positioned at the first position; and the second shock absorbing post is positioned at the second position.
4. The electromechanical chime mechanism of claim 1, wherein the mallet head is positioned relative to the chime bar such that the mallet head contacts the chime bar substantially at a center along a short dimension of the chime bar.
5. The electromechanical chime mechanism of claim 1, wherein the mallet head is positioned relative to the chime bar such that the mallet head contacts the chime bar substantially at a center

along a long dimension of the chime bar.

6. The electromechanical chime mechanism of claim 1, wherein the chime bar is configured to, when struck by the mallet head, output sound at a predetermined frequency that is lower than a natural frequency of a bar that (i) is composed of a material having a substantially similar density and a substantially similar elasticity as a material of the chime bar and (ii) has the same outer dimensions as the chime bar.

7. The electromechanical chime mechanism of claim 6, wherein the chime bar has outer dimensions that are no greater than 3.1 mm, 147 mm, and 38 mm.

8. The electromechanical chime mechanism of claim 6, wherein the chime bar is composed of a material having a density no greater than 2.8 g/cm³.

9. The electromechanical chime mechanism of claim 6, wherein the predetermined frequency substantially corresponds to a B4 note.

10. The electromechanical chime mechanism of claim 1, further comprising a plurality of shock-absorbing elastic posts with crescent tops holding the chime bar in place.

11. The electromechanical chime mechanism of claim 10, wherein the shock-absorbing elastic posts are configured to hold the chime bar in place while allowing the chime bar to vibrate freely.

12. The electromechanical chime mechanism of claim 1, wherein the chime bar has a rounded rectangular shape.

13. A rectangular chime bar comprising: a plurality of rectangular undercuts, wherein: the chime bar is configured to output sound at a predetermined frequency in response to being struck; a fundamental vibrational mode of the chime bar corresponds to the predetermined frequency; a first position along a long dimension of the chime bar corresponds to a first node of a fundamental frequency and a second position along the long dimension of the chime bar corresponds to a second node of the fundamental frequency; and the plurality of rectangular undercuts includes a first undercut located at the first position and a second undercut located at the second position.

14. The rectangular chime bar of claim 13, wherein: the predetermined frequency is lower than a natural frequency of a bar composed of a same material and having same outer dimensions as the chime bar.
